

ing hub is because of the availability of comparatively cheaper, skilled manpower, particularly experts in semiconductor and embedded engineering. Apart from that, there is untapped potential in rural areas where electronics and related technologies have not penetrated much. This opens up immense possibilities.

Lack of advanced technology and infrastructure

Lack of capability to assemble new surface mount packages and limitation on density handling due to a wide process window index (PWI) in the manufacturing process are two challenges faced by Indian SMT industry, which directly affect the manufacturing yield. Because of the above reasons, equipment having higher cycle per hour (CPH) capability and lower placement distance ratio are not being introduced. Other challenges include high cost of capital equipment and raw materials, non-availability of finance and lack of a favourable infrastructure towards the growth of SMT industry.

Boost from the government

The government of India has been encouraging the development of special-purpose machines (SPMs) for use in electronics manufacturing, and in specific for surface mount component assemblies. The government's Department of Electronics and Information Technology (DeiTY) has come up with many promising policies for the development of Indian electronics industry. It has laid high emphasis on indigenous electronics manufacturing, which is being promoted through several incentives, including schemes for enhancing set-ups using capital equipment. "This will, for sure, bring in advanced manufacturing equipment aimed at serving the domestic industry," believes Ankan Mitra.

Though there are no policies specifically targeting the SMT industry, existing policies, like national policy on electronics (NPE), national manufacturing policy (NMP), modified special incentive package scheme (M-SIPS) and electronic manufacturing clusters (EMCs), indirectly benefit the SMT sector. A few other

MAJOR CONTRIBUTORS TO THIS REPORT



A. Vijayendra
managing director,
IPC India



Anilkumar Paramalla
director, Hermes Electronics
Pvt Ltd



Ankan Mitra
vice president,
SMTA India chapter



David W. Bergman
vice president for
international relations, IPC

schemes have also been introduced last year, and these look promising too.

Make in India movement. This campaign launched in September last year, with an aim to attract investment and turn India into a major global manufacturing hub, has received a lot of attention and support in the country and international markets. It has helped bring about major enhancement in the Indian electronics industry.

Electronics development fund (EDF). In December 2014, the cabinet approved EDF for promoting new ventures, especially in electronics system design and manufacturing (ESDM), nanoelectronics and IT sectors, to achieve net zero imports by 2020. This scheme is meant to provide financial aid for developing international-class infrastructure for Indian electronics industry. This boost for ESDM is going to affect SMT equipment industry too.

The fault in our laws

Though there seem to be many encouraging ventures, from central as well as various state governments, with an aim to encourage electronics manufacturing, the most important factor is the implementation of these policies effectively, says Anilkumar, director, Hermes Electronics Pvt Ltd. "Unfortunately these incentives and policies mostly do not reach, and are not accessible to, local entrepreneurs or promoters," he adds.

According to industry experts from ELCINA Electronic Industries Association of India, tax irritants, such as special additional duty (SAD) and central

sales tax (CST), anomalies in tax policies and complexity of government procedures, are what hamper investment in India. Implementing policy corrections would help establish a conducive ecosystem for electronics manufacturing and associated businesses.

Supporting organisations, courses and certifications

Surface Mount Technology Association (SMTA) is an international organisation for professionals in electronics assembly technologies and related business operations. The India chapter of SMTA was inaugurated last year. Apart from providing support for SMT personnel, they conduct online training courses, like stencil printing and troubleshooting. SMTA also provides SMT processes certification and six sigma greenbelt courses, though these are currently not available in India.

Standardisation Testing and Quality Certification (STQC) directorate of DeiTY provides hands-on training courses in SMT for production, quality assurance and R&D personnel. These give engineers or technicians the knowledge and skill to effectively perform material selection (for solder paste, cleansing agents, etc), hand soldering, wave soldering, reflow soldering, inspection, rework and repair of surface mount assemblies.

IPC is a global association connecting the electronics industry and generating standards for the same. It offers training and certification programmes in India, which can be taken up by professionals from the SMT field. It also supports SMTA India activities. ●